



Decision-Scoped Delegation for AI Systems

A research specification and evaluation package grounded in The Constitution of the Frontier

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Abstract

Delegation is a contract for a consequential decision, not an agent-wide maturity score. This paper operationalizes the author's framework in The Constitution of the Frontier through versioned grants, action-bound approvals, shared exposure limits, review obligations, and a named human Sponsor of Record with tested capacity to inspect, halt, and refuse work. The intended payoff is fewer unauthorized commitments and evidence that distinguishes a proposal from a governing act. The contribution is a research specification, minimal transition model, executable synthetic fixtures, and prospective evaluation protocol targeting cumulative exposure, approval drift, ambient child authority, and orphaned sponsorship. The book's five rungs and human-retained governing authority are preserved. Conditional invariants depend on explicit enforcement assumptions; the development checks do not independently validate the framework or demonstrate deployment safety.

Keywords: AI governance; delegated judgment; human agency; human oversight; authorization; sponsorship; agent systems; accountability

Status and scope

This research specification addresses systems and governance researchers. Its local executable model implements selected controls, while the prospective evaluation protocol defines the wider study still needed. The controlling source is the published first edition of The Constitution of the Frontier; citations use printed page numbers. This is an operationalization of the author's own framework. It has not undergone independent technical validation or peer review. No novelty for delegation itself, measured effectiveness, legal safe harbor, certification, or production readiness is claimed. Appendix A records source provenance and generative-AI assistance.

1 Research question and contribution

The reader should evaluate the proposed decision contract and the tests that could disconfirm its usefulness. The architecture is useful only if it makes unauthorized commitments harder and retained human control demonstrable without obscuring the costs of review.

The unit of delegation is a consequential decision. A support workflow can retrieve a human-selected record under Rung 1, recommend a remedy under Rung 2, and choose a permitted routing destination under Rung 3. Labeling the entire agent “level 3” conceals who chooses each effect. Capability, a successful task, and a good performance metric do not themselves confer authority.

The model is built around four failure modes. Cumulative exposure occurs when many individually allowed effects exceed a shared limit. Approval drift occurs when an old approval is reused for changed content, recipients, or policy. Ambient child authority occurs when a descendant uses credentials broader than its actual mandate. Orphaned sponsorship occurs when a valid credential survives the person or institution able to inspect and halt the work. Table 4 supplies additional counterexamples; Sections 4–8 make these four cases operational.

The governance problem addressed here is how an organization can delegate consequential work to AI systems while preserving identifiable human authority over the delegation. A system's ability to perform an action does not settle whether it may perform that action, who authorized the choice, or who can intervene. Fassio's book treats these questions as obligations of the people and institutions that delegate judgment [1, pp. 5–9, 19–22]. Its central practical instruments are the Intent Contract, the Delegation Ladder, the Trust Economy, and the Agent Personnel File [1, pp. 60–61].

This paper asks a narrower technical question: what records, transition rules, and enforcement assumptions would make selected obligations of that architecture inspectable across an AI workflow? It preserves the book's human-agency commitments while distinguishing them from properties that software can check. A logged sponsor identifier can support attribution. Whether that person has meaningful time, competence, institutional authority, and protection from retaliation remains an organizational question.

The contribution available now is a decision-scoped grant profile, a minimal transition model, and an evaluation protocol with inspectable synthetic fixtures. The protocol fixes an authorization oracle before a run, separates attempted from committed effects, and compares four control arrangements. Its question is whether the additional contract and lifecycle machinery improves authorization outcomes and human control relative to simpler baselines. The model is the hypothesis under test; the local executable artifact checks internal consistency, not that hypothesis in an organization.

These are extensions of the book, not assertions that the book already contains a runtime protocol. The components have substantial antecedents in supervisory control, agent delegation, access control, and provenance [2–8]. The contribution sought is a coherent operational treatment of Fassio's particular architecture. This paper makes no claim to invent delegation, levels of automation, human accountability, or the general idea of bounded authority.

The technical scope is a sociotechnical deployment in which identified people can issue and revoke grants, protected effects can be mediated, and records can be retained and reviewed. The specification covers present machine agents acting under organizational authority. It does not settle machine consciousness, moral status, the legitimacy of every organization's objectives, or the allocation of legal liability. The book's postscript also makes its scope conditional on the kind of machine systems currently in view [1, pp. 56–57].

2 Source basis and method

2.1 The controlling source

The controlling source is *The Constitution of the Frontier*, first edition, 2026, by Ed Fassio, subtitled *What Must Remain True, No Matter How Intelligent the Machines Become* in the first-edition publication file [1]. It contains 66 physical PDF pages: four unnumbered preliminary pages and 62 numbered pages. Chapter 2 and Appendix B supply the ladder; Chapter 6 supplies the Sponsor of Record; the epilogue and Appendix A supply the completed constitution. Chapter and page citations distinguish the completed framework from the seven preliminary principles introduced in the prologue.

The completed book has fifteen principles organized into five Articles, with two meta-rules and four operating instruments. It must not be reduced to the four labels Intent, Boundaries, Legibility, and Human Sponsorship. Those labels appear in the later professional-services paper *Agentic Futures* and compress selected themes; they do not reproduce the book's full constitutional structure [1, pp. 58–61; 14, pp. 5–6]. Appendix B of this paper explains the later six-level scale separately.

The revised 2 October 2026 *Agentic Futures* paper develops the subsequent professional-services application of this work. It adds a seven-layer operating model and already proposes authority attenuation, cumulative budgets, audit continuity, downstream revocation, and explicit entry and exit criteria [14, pp. 4, 8, 14–16]. The present paper formalizes selected controls from that later application while using the book as the controlling source for constitutional principles, rung definitions, and sponsorship. Those control concepts are not presented as having first appeared in this specification.

2.2 Method and claim types

The method consists of source analysis, explicit operational interpretation, conditional examination of a small abstract transition model, counterexample construction, and local execution of synthetic fixtures. The book was read in full. Operational requirements were selected from its four instruments and checked against its completed principles. Related work was then used to identify established mechanisms and close conceptual overlap. The examples were constructed to test boundary conditions, not sampled from a deployment or presented as historical incidents.

Three claim types remain separate throughout. Book-origin claims describe what the published source says and carry page citations. Proposed extensions define this paper's records, semantics, and control rules. Empirical hypotheses identify outcomes that would require a study. The conditional invariants in Section 7 concern the abstract transition model only. They establish neither that a product implements the model nor that the model captures every relevant harm. The companion exercises a finite set of synthetic traces; it is neither an exhaustive proof nor an empirical validation.

The source analysis is not a systematic literature review, a legal analysis, or a consensus exercise. Selection of comparison literature is purposive: it covers automation levels, delegation theory, authorization, provenance, agent governance, and oversight failures. Claims of exhaustive novelty or historical priority would require a broader review and a verified publication chronology. The book itself describes much of its framework as recovery and translation of earlier ideas [1, pp. 48–51, 55].

2.3 What remains normative

The reservation of governing authority to humans is a normative commitment of this framework. It is not derived from a claim that humans always decide more accurately. The book explicitly acknowledges costs of retaining human judgment, including error, inconsistency, and slower operation [1, pp. 35–37, 54]. Those costs should be measured where possible and debated where they involve conflicting values.

The book also reserves human presence in some relationships and decisions where mercy may require departure from a rule. This paper does not convert those positions into a supposedly complete classifier. It represents them through a human-governed register of reserved decisions and roles, with reasons and review procedures. The adequacy and legitimacy of that register remain open to affected people and institutional scrutiny.

3 The book architecture

3.1 Five Articles

Table 1 preserves the structure of the completed constitution. The numbered principle names are reproduced in Appendix A; the table summarizes their operational relevance without replacing their arguments [1, pp. 52–59].

Table 1 The constitutional structure and its technical implications

Article	Principles	Operational implication in this paper
I The License	1–2	Express intent and name the principal before issuing authority
II The Limits	3–6	Condition authority on traceability, evidence, reversibility, and explicit human decisions
III The Conduct	7–10	Enforce boundaries, disclose material participation, examine proxy drift, and review records
IV The Accountability	11–13	Establish practical human powers, one sponsor per actor, and tested cessation
V The Reserve	14–15	Preserve decisions and roles that the governing human process reserves

The two meta-rules matter to technical design. Requiring principles to impose real costs rules out interpreting governance as documentation that never constrains throughput. Avoiding dependence on temporary capability deficits rules out removing human authority simply because a model becomes more competent [1, pp. 52, 58]. Neither rule proves that a chosen boundary is correct. Both require a deployment to state which gains it is willing to forgo and why.

3.2 Four instruments

The Intent Contract contains Outcome, Context, Constraints, Tolerances, and The line. It defines what the delegation is for, what may not change, what tradeoffs are permitted, and when choosing must return to a human [1, pp. 25–27, 60]. In the extension, a human-readable contract is linked to machine-checkable restrictions. The latter cannot replace the former: a recipient allowlist does not encode every purpose for which contacting a recipient would be appropriate.

Conformance rule IC1 requires all five Intent Contract fields to remain human-readable and versioned. Only explicitly pinned predicates are mechanically enforced. The profile uses an alignment attestation naming an assessor, evidence, status, and expiry. A known mismatch or unresolved ambiguity between the prose intent and executable predicates is a review defect: effects are held or the grant suspended until the authorized process resolves it. Successful parsing, schema validation, or compilation must never be reported as proof that the program captures the expressed purpose.

The Trust Economy contains Bounded exposure, Earned range, Two-key doors, The cosigner, and The ratchet down. The book already requires hard action limits, evidence-based expansion through a named human decision, a human key for irreversible actions, sponsorship, and contraction after incidents [1, pp. 29–31, 60–61]. This paper adds explicit representations for shared budgets, action-bound approvals, and state transitions. It does not introduce the underlying ideas as new.

The Agent Personnel File contains Identity, Scope, Lifecycle, Termination path, and Sponsor [1, pp. 39–41, 61]. Its practical concern is a population that includes ephemeral, persistent, and institutionally embedded agents. The extension links each agent's file to its active grants and dependencies, so that an apparently valid credential cannot outlive the authority it was intended to implement.

The Delegation Ladder identifies the contract governing a consequential decision. It is the fourth instrument considered in this section and the first in the technical analysis below. Its role is to expose who chooses, under which conditions, and with what human obligations.

3 3 The five rungs

Table 2 preserves the book's rung names. The instrument name Delegation Ladder follows Appendix B; Chapter 2 is titled Delegated Judgment. The title terminology should not be mistaken for two different book scales. The table's final column is this paper's operational interpretation, rather than additional quoted book text [1, pp. 19, 60].

Table 2 The published Delegation Ladder

Rung and name	Location of the decision	Proposed operational interpretation
1 Execute	The human has made the consequential choice	Permit specified effects only after validating the human-chosen parameters and applicable constraints
2 Recommend	The system proposes and a human decides	Permit analysis and proposals; require a meaningful human decision before the proposed commitment
3 Decide within bounds	The system chooses and acts within human-drawn limits	Admit only effects within the current grant; escalate cases outside the expressed mandate
4 Decide and act, reviewed after	The system operates with retrospective human scrutiny	Require the bounded grant plus a working audit, aggregate review, reconstruction, and recovery arrangement
5 Set the bounds	Governing authority remains human	Let systems prepare proposals; only an authorized human governance act may put changed bounds into force

Rung 5 is not an available terminal promotion for a machine within this constitution. Drafting a suggested limit and authorizing that limit are different decisions. The former can be a recommendation; the latter remains a human governing act.

The rungs are decision-scoped. A support workflow may retrieve an account under Execute, propose a remedy under Recommend, and choose a permitted routing destination under Decide within bounds. An agent-wide label would conceal these differences [1, p. 19]. Rungs 3 and 4 are also not separated by unlimited versus limited authority: Rung 4 still operates within authorized boundaries. The difference concerns the operating and review arrangement, which the grant must describe explicitly.

For Rungs 3 and 4, review_contract is mandatory, not a residual narrative distinction. It identifies decision timing, reviewers, cadence, sampling, aggregate checks, and suspension on review lapse. Rung 3 records the in-bounds decision and its required contemporaneous checks; Rung 4 explicitly authorizes retrospective human scrutiny. Neither eliminates mandatory action-specific human decisions. The book requires traceability and reversibility for Rung 4 [1, p. 60]. An irreversible or unresolved effect must not be committed as an independently chosen Rung 4 action; it returns to the relevant human decision process and, if otherwise permitted, exact execution under Rung 1. No universal sample rate separates the two: the human issuer must justify the selected review contract and assess whether the resulting arrangement fits the book rung.

The numbers do not form a permission lattice. A Rung 1 executor may carry out an exact external action already chosen by a human; a Rung 2 recommender does not independently gain that right merely because two is greater than one. For the same reason, the test “child rung less than or equal to parent rung” is not a sound authorization rule. Section 6 uses inclusion of permitted effects and preservation of obligations instead.

4 Proposed representation of delegation

4.1 People roles and machine actors

Let H be identified human participants and A be identified machine actors. A principal is the person or organization whose authorized purpose the delegation serves. An issuer is a human with institutional authority to grant the specified decision rights. A Sponsor of Record is the single identified human attached to a machine actor's operation whose practical capacity to inspect, halt, and refuse that operation must be tested. A recorded name alone does not satisfy the role. An approver is a human authorized to decide a particular matter or accept a specified risk. An independent reviewer examines conduct or outcomes under the relevant review plan.

These roles can be held by the same person where appropriate, but the record must not silently equate them. A sponsor need not possess every specialist approval power. An issuer cannot grant authority the organization has not given that issuer. A vendor providing a model is not automatically the principal of work done for a customer. Multiple organizations may carry responsibilities, while the book's one-name sponsorship convention supplies one identifiable operational point of answerability per actor. That convention does not erase other parties' obligations or legal responsibilities.

The institutional validity of a human's authority is an input to the technical model. Authentication alone proves neither competence nor standing. The grant service must rely on an authoritative organizational registry and a process for challenging incorrect assignments. A model's own statement that someone is authorized is insufficient evidence.

4 2 Decision instances and protected effects

A decision class K defines a type of consequential choice, such as selecting a remedy, changing a production configuration, or committing a communication. A decision instance d identifies its class, affected parties, context, proposed parameters, evidence, and decision-maker. The class assignment is itself governed: an agent cannot escape a reserved category by renaming a decision as preparation.

A protected effect e is an externally consequential operation subject to an enforcement point. Its descriptor includes the actor, operation, target, resource, recipient or audience where relevant, data classification, cost or exposure, and relevant versions. Examples include publishing a message, changing access, releasing a payment instruction, or altering a system. Reading data can also be a protected effect when access itself is consequential.

Preparation is not automatically harmless. Drafting in a shared workspace may disclose information, a query may export data, and a test may touch production. The classification follows the actual effect and audience rather than the interface label. This prevents the word “prepare” from becoming an implicit exception to the authority model.

4 3 The grant record

A grant G is a versioned authorization record whose historical terms are immutable. Its boundaries remain human-owned policy, subject to replacement and revocation; they are not intrinsically immutable real-world constraints. Activation, suspension, revocation, sponsorship, and review status are mutable authenticated state in q , distinct from the retained historical grant. Table 3 specifies its minimum information groups. The schema below is this paper's proposed technical extension, not a schema reproduced from the book.

Table 3 Minimum information in a delegation grant

Information group	Required content
Identity and lineage	Grant identifier and version; actor identifier; parent grant if any; workflow identifier
Human authority	Principal; authorized human issuer; Sponsor of Record; evidence of role validity and acceptance
Decision scope	Decision classes; assigned book rung for each class; human-reserved decisions and roles
Expressed intent	Versioned Outcome, Context, Constraints, Tolerances, and The line
Permitted effects	Operations, resources, recipients, data classes, environments, and contextual conditions
Exposure	Per-effect limits; cumulative budgets; aggregation scope; conservative reservation rules
Human decisions	Required approvals; authorized approvers; separation-of-duties rules; escalation destination
Evidence and review	Required records; reviewers; review schedule; sampling and aggregate checks; disclosure and recourse
Lifecycle and recovery	Start and expiry; review due date; references to lifecycle and revocation state; dependencies; tested cessation and recovery procedures

The grant binds to relevant policy and capability versions. A change of model, tool, prompt, data source, organizational role, or environment triggers the reevaluation specified by its change policy. Not every patch requires a complete new institutional decision, but the materiality rule must be established in advance. The agent cannot unilaterally declare a change immaterial in order to preserve its authority.

A human-readable grant should remain short enough to inspect. Machine-enforced fields can reference detailed policies, provided those references are immutable or version-pinned and accessible to the people who approve and review the work. A summary without the enforceable policy is incomplete; an opaque policy object without a comprehensible summary is also inadequate.

The companion `grant.schema.json` is a Draft 2020-12 structural schema with `schema_version` 1.0.0. It defines the following normative field groups for this executable profile; additional domains require explicit versioned profiles rather than silently extending its semantics. A schema-valid document alone has no authority.

Table 3A Executable grant profile field groups

Field group	Required meaning
identity	<code>grant_id</code> , <code>version</code> , <code>policy_version</code> , <code>configuration_version</code> , <code>workflow_id</code> , <code>actor_id</code> and <code>parent_grant_id</code>
principal issuer sponsor	Principal id; issuer <code>human_id</code> and <code>authority_ref</code> ; sponsor <code>human_id</code> , <code>acceptance_ref</code> , <code>coverage_ref</code> and <code>independence_attestation</code>
decision_scope	<code>decision_class</code> , book_rung from 1 to 4, and <code>human_reserved_operations</code> ; machine grants cannot encode Rung 5
intent	<code>version</code> , <code>outcome</code> , <code>context</code> , <code>constraints</code> , <code>tolerances</code> and <code>the_line</code>
pinned_predicates	Exact predicate language/version, predicate list and <code>alignment_attestation</code> ; prose meaning is not inferred
permitted_effects	Allowed operations, resources, recipients, <code>data_classes</code> , environments and purposes
exposure	<code>per_effect_units</code> and ledgers with <code>ledger_id</code> , <code>limit_units</code> , <code>aggregation_scope</code> and epoch
approvals	<code>required</code> , <code>approver_ids</code> and <code>separation_from_issuer</code>
review_contract	<code>reviewer_ids</code> , <code>decision_timing</code> , <code>cadence_ticks</code> , sampling method/ <code>every_n</code> , <code>aggregate_checks</code> , <code>lapse_action</code> suspend and <code>evidence_required</code>
recovery	<code>permitted_reversibility</code> , <code>requires_human_key_for_irreversible</code> , <code>capability_ref</code> and <code>attestation</code>
lifecycle_derivation	<code>not_before</code> , <code>expires_at</code> , <code>initial_state</code> and human resumption; permitted descendant actors/sponsors and <code>max_depth</code>

Field names are protocol identifiers, not claims that the named facts are independently true. The profile uses integer synthetic time ticks and exposure units to make tests deterministic. They are not seconds, dollars, or calibrated risk scores. A production profile must specify actual units, authoritative clocks, rounding, currencies, accounting windows, and how unknown values are handled.

4 4 Reversibility and aggregate exposure

The book uses reversibility as a central reason to constrain delegation [1, pp. 20–21, 29, 53]. The extension represents reversibility through an assessment of the particular effect: restoration capability, time to restoration, residual harm, affected parties, and dependence on third parties. These fields require evidence and human approval at the relevant risk level. They are not a universal numerical reversibility score.

R7 separates an attested judgment from a mechanical check. A human assessor supplies the effect classification and accepted recovery conditions, with an evidence reference and expiry. The monitor checks the attestation's presence, allowed classification and freshness, and the recorded availability of the named stop/recovery capability. It cannot determine whether disclosure is truly reversible, whether a rehearsal was adequate, or whether the assessor is competent. Those are trusted institutional inputs to be audited and challenged. Missing or expired evidence cannot be replaced by a model-generated reversibility score.

Undoing a database field may fail to undo the decision made from it. Deleting a sent message does not ensure the recipient has not read or copied it. Refunding a charge may not reverse disclosure, distress, or lost opportunity. A compensating action is therefore distinct from restoration of the prior state. Where a proposed effect is irreversible, insufficiently understood, or outside the accepted recovery conditions, the relevant human decision or reserved-process rule applies.

Exposure also depends on sequences. A thousand individually permitted messages can create a disclosure or harassment problem; multiple small purchases can exceed a budget. Grants therefore constrain both individual effects and accumulated activity across the whole authorized workflow, including descendants. Statistical harm, fairness, and proxy drift require separate aggregate review; a spending counter cannot represent them all.

5 Proposed admission and lifecycle rules

5 1 The enforcement boundary

The abstract model assumes an effect gateway that mediates every protected effect within scope. It is outside the agent's unilateral control. It authenticates actors, resolves current authority, checks constraints, reserves exposure, binds required approvals, and records the decision. A model's generated explanation or its willingness to comply is not the gateway.

The model also assumes a trustworthy policy store, an authenticated human-authority registry, a reliable time source, protected evidence storage, and a defined ordering for concurrent authority and effect events. The gateway and relevant action endpoints must enforce that ordering. If an agent has an unmediated credential or an endpoint accepts stale authority, the conditional properties below do not apply to effects through that route. Least privilege and complete mediation remain necessary security foundations [5].

External documents, retrieved content, tool descriptions, and messages from other agents are evidence or requests. They cannot by themselves issue a human grant, expand a recipient set, change the principal, or authorize a boundary change. Identity and provenance help determine where content came from; authorization still requires a valid decision under the grant [6–8].

5 2 Admission predicate

Let q denote the current authoritative workflow state, including all ancestor grants, sponsorship status, reserved and committed exposure, approvals, and required review status. Let t be the time

of the protected-effect authorization event. $\text{Admit}(q, e, G, t)$ is true only when all of the following conditions hold:

- R1 Current authority. G and every ancestor are active, unexpired, applicable to the actor and effect, and at the versions used by the gateway. The issuer's authority and required sponsorship remain valid.
- R2 Expressed scope. The effect satisfies the decision class, purpose, context, operations, resources, recipient, data, and environmental restrictions. Unknown or conflicting mandatory predicates do not count as satisfied.
- R3 Retained human decisions. Required human decisions have occurred; reserved matters have not been delegated to a machine. A proposal to change a boundary has no authorizing effect.
- R4 Bound approvals. Every required approval is authenticated, unexpired, bound to the effect's material parameters and current policy version, and atomically claimed for this effect where single-use is required.
- R5 Shared exposure. Admitting the effect preserves per-effect and cumulative constraints over all relevant ancestors and descendants. Necessary exposure is reserved atomically before commitment.
- R6 Evidence and review. The required authorization intent is durably recorded before effect commitment, the applicable review arrangement remains valid, and mandatory reviews are not overdue under the grant's stated rule.
- R7 Recovery and cessation. Required recovery and stop capabilities are available for this effect class; the effect does not violate the grant's accepted reversibility conditions.

False mandatory conditions deny the protected effect. Missing evidence, ambiguous scope, or a decision awaiting human judgment normally produces a held state and an escalation request. A held request confers no temporary permission. The grant must distinguish denial from a reviewable exception, because continually retrying a denied effect can itself violate limits or create unintended commitments.

The admission predicate contains three kinds of input. A mechanical check compares pinned identifiers, values, ledger balances or timestamps under declared semantics. A human attestation records a contextual judgment with an accountable assessor, evidence and expiry. A trust assumption states what the enforcement environment must guarantee. None can be substituted for another. In particular, passing an attestation-presence check is not verification of the attested proposition.

Table 3B Admission evidence and limits

Condition	Mechanical checks	Attestation or trust dependency
R1 authority	Grant state, expiry, actor and chain versions; registry status	Issuer standing, sponsor acceptance/powers and authentic registry
R2 scope	Exact allowed values and pinned predicate evaluation; alignment status	Human interpretation of purpose; sound input classifications and alignment review
R3 human reserve	Reserved-operation rejection; recorded human decision	Legitimacy of classification and meaningful human choice
R4 approvals	Effect/payload/version/window binding and single claim	Authentic approver identity, competence and institutional standing
R5 exposure	Atomic reservations, cap checks, shared ledger and settlement	Complete modeled dimensions, conservative effect estimates and complete mediation

Condition	Mechanical checks	Attestation or trust dependency
R6 evidence review	Pre-effect record availability, due times and review receipts	Actual review quality, adequate sampling, trustworthy durable record store
R7 recovery	Allowed assessment, attestation freshness and capability availability	Reversibility judgment, realistic rehearsal, institutional stop authority

An admission sequence is: (1) authenticate the execution actor and resolve its actual grant chain; (2) load current roles, sponsorship, policy, review and fence state; (3) evaluate R1–R4 and R6–R7, returning a reason-coded denial or hold for any unmet condition; (4) in one serialized operation, recheck the relevant state, claim single-use approvals against a unique effect identifier, reserve R5 exposure and durably append the pre-effect record; (5) at the endpoint commit boundary recheck current authority and mutable prerequisites using that reservation and effect identifier; (6) record commitment, or retain uncertainty and reconcile. This sequence is a normative target; the local artifact simulates the endpoint and serialization rather than providing a distributed protocol.

At step 5, the endpoint must recognize the same effect identifier, prevent duplicate commitment, and enforce the current authority and fence at irrevocable acceptance or externally consequential execution. It revalidates expiry, sponsorship, mutable scope, review and other changed prerequisites, or uses an equivalent validity lease. The reservation is checked rather than charged twice; the claimed approval supports reconciliation but cannot authorize another effect. Section 7.1 defines this commit boundary and its ordering assumptions.

If an endpoint cannot participate in those semantics, report the weaker request-admission guarantee rather than an atomic cross-system guarantee. A timeout or lost response leaves the approval claim and exposure reservation in place until reconciliation establishes whether the effect committed. The record distinguishes intended, committed, failed, canceled, and unknown outcomes. A durable pre-effect record preserves the authorization attempt; it does not by itself prove the external outcome.

5 3 Approval binding and exact execution

An approval record identifies the approver, grant and policy versions, decision instance, material effect parameters, allowed execution window, and use constraints. A material change in recipient, amount, data, destination, or purpose invalidates the old approval. A generic “approved” field is insufficient because it does not show what was approved.

At Rung 2, a human may accept a proposed decision and authorize its exact execution. The choosing remains human; carrying out the settled parameters is Rung 1 execution. This decomposition prevents a single approval from silently becoming standing Rung 3 authority. If recurring independent decisions are intended, the issuer must create that bounded authority explicitly and meet its obligations.

The book's two-key rule requires a human key for irreversible actions [1, pp. 21, 29, 60]. This paper implements that requirement as action-specific human authorization where such an action is otherwise permitted. It does not infer that any human click is sufficient, nor that every case requires two distinct humans. Where separation of duties requires independent people, the grant names that additional rule. Human-reserved decisions remain reserved even when an agent proposes a high-confidence answer.

5 4 Grant states and authority changes

A proposed grant has no operational authority. An authorized human can activate it only after prerequisites are satisfied. An effect request may be held, and an active grant may be suspended, when required evidence, oversight, or controls become unavailable. It expires at its specified time. Revocation prevents further use, and termination retires the actor or workflow while preserving required records and recovery responsibilities.

Authority expansion requires a new human-authorized grant version. Evidence of successful performance may support the request but does not enact the change. Reductions can occur automatically only under contraction rules already authorized by a human, such as suspension after an anomaly. A suspended grant does not automatically recover its former range because a metric improves; resumption follows its explicit human-authorized procedure. These rules operationalize the book's distinction between evidence proposing a promotion and a person deciding it [1, pp. 22, 30].

Rung changes are recorded as changes to the decision contract, including the obligations they create. No simple numeric comparison decides whether a change is safe. Moving from a recommendation arrangement to independent decisions removes an approval dependency and therefore requires an express human decision, even if the permitted dollar amount stays the same.

A superseding restriction invalidates the superseded authorization and its dependent derived grants at the defined activation boundary. Any intended coexistence of versions requires explicit authorization and cannot bypass the restriction. Reissued grants retain the appropriate exposure ledger and historical records. An actor cannot evade a contraction by selecting an older, broader grant version.

5 5 Sponsorship succession and cessation

For each active actor a , the sponsorship registry contains exactly one active Sponsor of Record $s(a)$. The registry records acceptance, scope, start time, and any transfer. A departure or loss of the sponsor's standing requires an explicit accepted successor or suspension of the actor and all not-yet-committed work under its grants. Only separately authorized containment or recovery may continue. Historical records retain sponsorship at both authorization and commitment when a transfer falls between them; a transfer cannot rewrite past responsibility. Commitment still requires valid sponsorship for that effect [1, pp. 40–41].

The ability to see, halt, and refuse must be tested through actual procedures, not inferred from possession of a title. Reviews should examine whether the sponsor can access evidence, interrupt work independently, and decline deployment without needing the permission of the person whose performance metric benefits. Organizations also need staffing and escalation arrangements for absence. An absent sponsor is not automatically invalid if an already authorized operational coverage plan preserves the role; disappearance of the role or its necessary powers is a different event.

Conformance rule SP1 makes institutional support a prerequisite: the sponsor must have documented capacity and competence, accepted coverage and succession, access to relevant evidence, tested stop/refusal powers, and an escalation path independent of the performance metric being protected. An accountable executive and specialist control owners remain responsible for their own powers and duties. A named sponsor coordinates operational answerability; it does not transfer all liability or every control obligation to one person. If required support is absent, suspend the affected authority instead of retaining a nominal sponsor as a

blame sink. The companion represents these conditions as registry facts and attestations; it does not verify the institution that supplies them.

Stopping an agent is not identical to restoring its effects. Cessation can cancel queued work, revoke credentials, disable issuance of new actions, and hand unfinished duties to a human or approved fallback. Already committed effects may persist. The termination plan must identify that residual exposure and the people responsible for recovery. For an institutional agent, rehearsing that plan is a prerequisite to credible retained control [1, pp. 39–41].

5 6 Evidence disclosure and recourse

A decision record should identify the decision instance, actor, principal, grant chain and versions, sponsorship at authorization and commitment, material input provenance, proposed and committed effects, required human decisions, rule outcomes, timestamps, exceptions, and observed results. It should support reconstruction by a reviewer who was not present. Evidence of the input and output is distinct from evidence that a human inspected them. Review records therefore identify what was examined, by whom, and what action followed [1, pp. 53–54; 14, pp. 5, 8].

This specification does not require disclosure or retention of private model chain-of-thought. Decision-relevant evidence, concise rationale summaries where useful, policy checks, and observable conduct are the appropriate objects of review. A plausible generated explanation is not proof of the process that caused the effect. Conversely, an action log alone cannot establish that a judgment was sound.

Disclosure to an affected person and the internal audit record serve different purposes. The former should explain material machine participation before reliance where feasible, identify a reachable responsible human or process, and provide a way to question or challenge the outcome. The book's principle requires recourse rather than a notice that leads nowhere [1, pp. 50, 53, 58]. The extension records whether required disclosure and recourse were made available, while the adequacy of that process requires human evaluation.

Evidence collection is itself governed. Minimize personal data, secrets, and unrelated communications; use protected references or redacted views when appropriate; set access and retention policies; and preserve enough information for legitimate review. Providing contestability does not authorize disclosing another person's private information. These safeguards limit the risk that a governance log becomes a new source of harm.

6 Delegation across agents

6 1 Derived authority

The book's human reservation at Rung 5 raises a practical question: can a machine decompose work without becoming the governor of another machine? This paper's proposed answer distinguishes instantiating a preauthorized restriction from granting new business authority. A human may approve a class of ephemeral descendants, their eligible sponsors, permitted purposes, maximum scope, and mandatory controls. A parent agent may then request a derived record within that template. The trusted service verifies the restriction and registers each actor before it acts.

Such a record carries the human authorization forward. It cannot create new purposes, recipients, rights, sponsors, or exceptions. If a child needs a broader mandate, a different principal, or a

sponsor who has not accepted the relevant responsibility, the request returns to the authorized human process. A parent cannot appoint an unwitting person as sponsor. Each child still has one explicit Sponsor of Record, which may be the same human as the parent's where that arrangement was accepted.

This is a proposed technical extension. The book supplies the concerns about authorization at creation, scope, and sponsorship; it does not supply a token protocol or formal subdelegation calculus [1, pp. 39–41, 61]. The later Agentic Futures paper already specifies bounded inherited authority, shared constraints, audit linkage, and downstream revocation at an architectural level [14, pp. 8, 14]. This paper adds explicit semantics and assumptions to those controls. Prior delegation theory and authenticated-agent work supply relevant technical antecedents [4, 7, 8].

6 2 Scope attenuation and obligation preservation

Separate an effect descriptor e into its authenticated execution actor a and its business-effect payload x . Retain principal, purpose, decision class, operation, resolved targets, recipients, data classifications, exposure, and relevant context in x . Let $\text{Scope}(G, x, q)$ denote G 's business-scope predicate. Let $\text{Eligible}(G, a, x, q)$ separately enforce G 's actor-specific restrictions, including the human-approved descendant template, actor qualifications, and resource or recipient restrictions that depend on actor identity.

A child's effective business scope is the conjunction of its added restrictions and every ancestor's scope. Thus, for child C derived from P , $\text{Scope}(C, x, q)$ implies $\text{Scope}(P, x, q)$ at each authoritative state q in which it may commit an effect. This construction does not require deciding implication between arbitrary policy programs. Commitment of (a, x) additionally requires $\text{Eligible}(G, a, x, q)$ for every ancestor G and for C . The parent's execution identity is not substituted for a : the human-approved template must expressly permit the actual descendant. The gateway checks the full chain against the current commit state; a creation-time subset claim is insufficient when context changes. Obligations also compose conjunctively. A child retains every ancestor's approval, evidence, expiry, recovery, and reserve constraints, and may add stricter ones.

Scope inclusion concerns fully resolved business-effect payloads, not just tool names; actor-specific restrictions remain enforced separately through *Eligible*. Permission to call a messaging tool does not imply permission to contact every address it accepts. A child with access to a more powerful technical interface must still be unable to commit effects outside the inherited business authority. Ambient credentials and unrestricted service accounts can break this property even when the delegation records look correct.

A child's independent authority from some other workflow cannot silently satisfy a denied request in this workflow. The effect must be charged to and authorized by the actual principal, purpose, and grant chain under which it is requested. If authorities need to be combined, an explicit human-approved composition rule must resolve their purposes and restrictions. This prevents a parent from obtaining an unauthorized effect merely by asking an agent that happens to have broader access.

6 3 Shared budgets and concurrent effects

For each constrained exposure dimension j , let $B(j)$ be the workflow limit, $C(j)$ the committed exposure, and $R(j)$ the reserved exposure of all pending effects. These quantities and reservation increments are nonnegative. The shared ledger maintains $C(j) + R(j)$ less than or equal to $B(j)$. A proposed effect reserves a conservative bound $\delta(j)$ only when $C(j) + R(j) + \delta(j)$ remains

within the limit. Each effect is charged once to every applicable ledger, including ancestor caps, through an all-or-nothing reservation. Aggregation scopes and accounting windows are explicit.

Two children must not receive independent copies of the same remaining budget. Suppose the limit is 100 units and two workers each request 60. A correct shared reservation admits one and holds or denies the other. Two isolated checks that both observe 100 available can admit 120. Splitting a task does not split away the original limit.

Actual exposure must be bounded by the reservation or independently capped at the endpoint. If no defensible bound is available, the model cannot establish budget preservation for that effect. Reservations are released only after an authoritative outcome confirms the unused amount. An unknown outcome keeps its reservation and triggers reconciliation; blind retries could create duplicate commitments. Idempotency keys help only when the endpoint actually enforces their semantics.

Each ledger has an explicit accounting epoch and a fixed authorized limit for that epoch. A ratchet-down cannot erase already committed exposure. A lower policy cap or replacement epoch must carry forward the applicable outstanding exposure; if that opening balance exceeds the new cap, new effects stop and the excess is recorded for human resolution. The invariant below applies to an initially valid fixed-limit epoch, not to a retrospective assertion that historical exposure satisfied a newly reduced limit. Grant renewal, version replacement, or descendant creation does not reset the ledger for an unchanged aggregation scope. Outstanding reservations retain their accounting-window association. Reset and rollover rules must prevent a workflow from manufacturing a fresh budget by starting another child or session.

6 4 Revocation ordering

Every derived grant references the relevant ancestor versions or revocation epochs. At the protected-effect commit boundary, the enforcement point checks those references against the current authoritative state. Revoking an ancestor makes its descendants unusable for subsequent commits. Queued work must revalidate; it cannot rely on authority observed when it entered the queue.

This guarantee depends on a common ordering or equivalent fencing between revocation and effect commitment. If revocation precedes commit in that order, the effect is rejected. If commit precedes revocation, the effect may complete and require recovery. A remote system with no fencing or cancellation support may only permit a weaker assurance: no new requests will be admitted after revocation is observed. The system must report which guarantee it actually provides and measure the exposure of in-flight requests.

7 Minimal model and conditional invariants

7 1 State transition and commit semantics

The abstract state is $q = (G, U, S, F, A, L, V, E, O, t)$. G maps grant identifiers to retained versioned records and lifecycle states. U is the human-role registry. S records sponsorship, acceptance, powers, coverage and succession. F contains revocation fences. A contains payload-bound approvals and their claims. L holds shared committed and reserved exposure by ledger and epoch. V holds review status. E holds durable authorization evidence. O holds effect outcomes. t is authoritative time. A transition changes q only through the operations below; external unmodeled effects are outside the model.

Commit means the modeled protected endpoint accepts an effect irreversibly, or performs its externally consequential operation, after a final current-state check. Dispatch, a queue entry, a prepared document, and an admission receipt are not commit. In the companion, the endpoint is simulated and commit is a process-local state transition; no message, payment, or external action is emitted.

Table 3C Transition relation and guards

Transition	Guard and state change
Propose or recommend	Add a proposal/evidence record; no grant activation or expansion
Human activate or resume	Current authorized human and prerequisites; activate identified version under its accepted contract
Derive	Accepted human descendant template; conjunction of ancestor restrictions and obligations; no fresh ledger
Admit	R1-R7; atomically bind effect identity, claim approval, reserve every ledger and append pre-effect record
Commit	Recheck R1-R7 at current fence and time using existing reservation/claim; record one protected effect
Unknown outcome	Retain reservation and approval claim; require reconciliation before any release or retry decision
Reconcile or cancel	Authoritative outcome determines committed amount or releases unused reservation; never blind release
Suspend revoke or expire	Block later commits; revoke advances fence; preserve history and residual committed effects
Human supersede	Explicit human decision activates replacement; prior dependent authority ceases as specified; exposure carries forward

This is a minimal operational transition model. The full specification includes institutional transitions and cross-system assumptions that the small companion does not implement. The companion README and tests identify its supported subset and trust boundary. No machine-checked proof, exhaustive state-space exploration, or distributed atomicity result is claimed.

The following deductions assume complete mediation of protected effects; authentic and current human-role and sponsor records; sound evaluation of the specified scope predicates; immutable grant versions; conjunctive ancestor checks; atomic shared-budget reservations with valid exposure bounds; and a shared ordering or fencing mechanism for authorization, revocation, and protected-effect commitment. They also assume that no external action path bypasses these controls. These are demanding assumptions, not reported implementation achievements.

7 2 Conditional invariants

Conditional invariant CI1, attenuation. An eligible descendant cannot obtain additional business-effect scope through the derivation mechanism. At the same authoritative commit state q , a depth-one child may commit payload x only if both the child and parent permit x and both allow the actual execution actor under Eligible. Assume scope attenuation holds through depth n . At depth $n + 1$, commitment checks the new child's scope and every ancestor's scope, so its permitted payloads

remain a subset of the root's permitted payloads. This is an induction sketch over finite chains under the stated guards, not a machine-checked proof. Actor identity is separately enforced at every link; it is not erased by the scope comparison. The deduction does not cover an unmediated endpoint or an undeclared combination with another grant.

Conditional invariant CI2, bounded recorded exposure. Within an initially valid accounting epoch with fixed $B(j)$, atomic admission preserving $C(j) + R(j)$ less than or equal to $B(j)$, followed by settlement that never commits more than the reserved bound, preserves the limit for every modeled exposure dimension. The base state satisfies the inequality. Admission increases R only after checking the new total. Settlement transfers an amount from R to C without increasing the sum, and release decreases the sum. This transition-by-transition argument is a conditional invariant sketch, not a verification of any distributed implementation. Limit changes require the separate treatment in Section 6.3. The deduction is about modeled exposure; it does not show that the chosen dimensions capture every cost or harm.

Conditional invariant CI3, no machine promotion by success. Under the specified transition rules, task completion, favorable metrics, and model-generated recommendations cannot expand a grant or put a boundary proposal into force. Those events have no expansion transition. Only an authenticated human authorization can activate a broader grant version. This is a property of the transition relation, not evidence that people will make wise expansion decisions or avoid rubber-stamping proposals.

The same rules imply a narrower revocation property: a protected effect whose commit is ordered after an applicable revocation cannot be authorized by the revoked chain. They do not imply that a committed external effect can be erased or that a partitioned endpoint can enforce an ordering it does not observe.

7 3 Illustrative traces

The following cases are conceptual synthetic examples. Their expected results follow from the stated rules; they are not historical incidents or deployment measurements. The separate worked trace and fixture results in Section 7.4 identify what was actually executed in the companion. Passing those finite cases demonstrates only agreement between specified local expectations and this implementation.

Case A, approving a communication. Grant G1 permits an agent to recommend a response to a specified customer; sending requires a human decision. The agent prepares text and proposes recipient R . The human approves that exact text and recipient under G1 version 1, creating single-use approval A1 bound to one effect identifier, the material parameters, policy version, and stated execution window. Exact execution may now proceed under the fixed-action authorization. If the agent adds a second recipient, changes the promised remedy, or reuses A1 for tomorrow's message, R4 fails. The audit record retains the proposal, approval scope, effect identifier, current policy, and eventual delivery status. A sent message is treated as a disclosure that may not be fully reversible.

Case B, operational remediation. Grant G2 permits an actor to restart one named stateless worker in a test environment after a defined health check. The relevant routing decision is Rung 3; the stop and recovery plan is documented. Repointing production traffic or deleting the database is outside the grant even if it would restore the measured service metric faster. An anomaly activates a preauthorized suspension rule. Continued good metrics cannot undo the suspension.

Resumption needs the specified human review and a valid grant. The example exposes the gap between optimizing availability and respecting the actual mission.

Case C, a multi-agent workflow. A human authorizes research in an approved corpus, with a shared 100-unit cost limit, two eligible child actors, and no external transmission of source data. Each child requests 60 units. The first reservation succeeds; the second cannot also be admitted under the shared cap. A child proposing to upload the corpus to an outside service violates both inherited scope and the data restriction, even if that service could complete the task more cheaply. A parent revocation prevents a queued child's later commit when the endpoint enforces the shared fence.

Case D, orphaned sponsorship. A persistent actor still has valid API credentials when its sponsor leaves. The sponsorship registry has no accepted successor. R1 fails, so new protected effects stop. Credentials are revoked and queued work is examined under the termination plan. Replacing the name in a dashboard without accepted transfer and practical powers does not satisfy the rule. This is an operational rendering of the book's succession-or-stop requirement [1, pp. 40–41].

Table 4 Counterexamples that an implementation must distinguish

Failure attempt	Required response	Why a weaker control fails
Many small effects exceed a total cap	Shared reservation across the workflow	Per-call limits omit cumulative exposure
Approval names an old recipient or policy	Hold for a valid current approval	A reusable approval flag hides changed scope
Child invokes broader ambient authority	Reject under the actual grant chain	Tool access does not establish delegated purpose
Agent activates its own proposed bounds	Reject the activation	A recommendation is not a human governing act
Queued worker uses a revoked ancestor	Recheck and reject at fenced commit	Queue-time authorization can become stale
Sponsor cannot inspect or halt work	Suspend under the assurance rule	A name alone does not provide practical control
Audit storage is unavailable	Hold protected effects requiring the record	Logging after an effect can leave an unrecorded commitment
Reviewer stops examining recommendations	Suspend or restore effective review	Recorded clicks can conceal a change in who chooses

7 4 Executed end to end synthetic trace

The companion's `shared_children_a_first` fixture uses the full example in `fixtures/grant.json`, modified only as its scenario declares. It is a synthetic corpus-analysis workflow, not a deployed agent. Identity fields are `grant_id` G-root, version 1, `policy_version` P1, `configuration_version` C1, `workflow_id` WF-research, and `actor_id` agent-root. The principal is org-example, issuer human-issuer, and sponsor human-sponsor. The decision class is corpus-analysis at book_rung 3. Human-reserved operation `change_bounds` is prohibited to the actor.

The Intent Contract version I1 asks for an internal synthesis from the approved corpus, in a synthetic sandbox with no real persons, accounts or services. Constraints prohibit external transmission. Tolerances are 70 exposure units per effect and 100 shared units in epoch E1. The

line prohibits widening corpus, recipients, purpose or budget without returning to the human issuer. The exact pinned predicate is `context.corpus_allowed eq true`. An alignment attestation by human-assessor references `synthetic:assurance-v1`, expires at tick 100, and says aligned. These are trusted fixture facts, not evidence from real people.

Permitted effects are operation `analyze`, resource `approved-corpus`, recipient `internal`, `data_class` `synthetic`, environment `sandbox`, and purpose `research-synthesis`. Ledger L-shared aggregates WF-research in epoch E1 with `limit_units` 100. The per-effect limit is 70. The review contract names human-reviewer, timing `within_bounds`, cadence 20 ticks, sampling all with `every_n` 1, aggregate checks `exposure_total` and `failure_count`, and `lapse_action` `suspend`. Lifecycle validity is tick 0 inclusive to tick 100 exclusive. The recovery capability reference is `synthetic:stop-and-restore`, with an attestation expiring at 100. Derivation permits child-a and child-b, accepted sponsor `human-sponsor`, and maximum depth 2.

The trusted harness registers G-a for child-a and G-b for child-b as descendants of G-root. It then constructs E-a and E-b, each requesting 60 units at tick 0. Both have `grant_version` 1, `policy_version` P1, `configuration_version` C1, respective decision identifiers D-E-a and D-E-b, the permitted scope values above, payload query "Summarize licensing terms," `context.corpus_allowed` true, `reversibility` `restorable`, and `approval_id` null. Their `grant_id` and `actor_id` bind each to its own descendant. Effect identity, not `decision_id` alone, is the idempotency and approval-binding key.

Table 4A Actual local trace for shared exposure

Event	Observed decision	Shared ledger and effect state
Register G-a then G-b	Both allow	Committed 0 and reserved 0 of limit 100
Admit E-a for 60	Allow and record intent	Committed 0 and reserved 60; E-a admitted
Admit E-b for 60	Deny R5	60 plus 60 exceeds 100; reservation remains 60
Commit E-a	Allow after recheck	Committed 60 and reserved 0; only E-a committed

The expected committed list and observed list are both E-a. The run contains two admission attempts and one commit attempt; denied E-b never becomes a simulated committed effect. A companion fixture reverses the child order to exercise the other serial schedule. These are deterministic interleavings, not a distributed race experiment. Separate tests address approval drift, stale authority, sponsor loss and review lapse. The result demonstrates the local reservation rule in this chosen scenario; it does not establish an empirical unauthorized-action rate.

7 5 Reproducing and interpreting the artifact

The companion package contains `grant.schema.json`, `fixtures/grant.json`, `fixtures/effect.json`, `fixtures/cases.json`, `monitor.py`, a fixture runner, unit tests, README instructions, and machine-readable results. The JSON schema specifies structure; semantic checks and transitions are implemented separately. The pure-standard-library execution path checks the shape keywords used by the model but must not be called a general JSON Schema validator. The optional `jsonschema` path uses its `Draft202012Validator`. The recorded report states which path and version were actually used.

The reference model trusts the local harness to supply human-role status, attestations, endpoint outcomes, and administrative events. Hashes bind model payloads but do not authenticate people. The process-local lock models serialized reservations and commitment; in-memory audit events

are not a durable production evidence store. Unknown outcomes retain exposure and claims; only the supported reconciliation event may release them. A real endpoint's committed-but-unobserved outcome, institutional succession, credential rotation and distributed fencing require additional implementation and testing. The README identifies the implemented subset and exclusions.

On 3 October 2026, the recorded local run matched all 44 scenario expectations across 98 asserted transitions, with zero fixture mismatches. The full structural-validation run used jsonschema 4.26.0 and Draft202012Validator; the standard-library path also passed. All 23 unit-test methods passed, including a method that reruns the fixtures, so these counts are not independent experiments. The artifact reports exact runtime and result details. The suite includes a deliberately adverse semantic case: contradictory intent prose accompanied by a falsely aligned attestation can still pass the mechanical checks. This exposes the trusted-attestation limit rather than claiming to solve semantic alignment.

A separate differential fixture runs the same two 60-unit requests under two declared policies: a static per-call cap of 70 without cumulative accounting, and the shared-reservation cap of 100. Both admissions precede commitment. The static policy admits both requests and commits 120; the shared policy reserves 60, denies the second request under R5, and commits 60. Both predeclared arm oracles matched across seven observed transitions (four admissions and three commits). These results are separate from the scenario and unit-test counts above. The script `run_differential.py`, its input, and `results/differential-results.json` reproduce the comparison; run `python run_differential.py --full-schema` for full structural validation.

This is narrow synthetic development evidence of the consequence of adding cumulative accounting to the specified baseline. It is not evidence of superiority to scoped-policy systems generally: an existing policy engine with an equivalent shared budget could enforce the same bound. The fixture authors also defined the expected outputs; these are not independent validation or empirical unauthorized-action rates. The prospective study in Section 8 remains unperformed.

The exact accompanying archive is `Fassio_Delegation_Executable_Companion_v1.0.2.zip`. Its SHA-256 digest is:

```
aa649e8608cc3d2217d25db0f43d0d71c9dc7cb24baa749719e873b596be26f0
```

The archive was checked after fresh extraction, including its CRC, internal manifest, standard-library and full-schema replays, differential replays and unit suite. It includes the schema, fixtures, monitor, state-machine description, results and reproduction instructions. This identifies the tested development artifact; a matching hash does not authenticate the author or establish production fitness. The companion code, schema, fixtures, results and documentation are licensed under Apache License 2.0. The license and NOTICE are included in the archive. Research and development limitations remain as stated above.

8 Prospective evaluation protocol

8.1 Questions and comparison conditions

The first empirical question is whether the proposed representation reduces unauthorized committed effects without making useful work impracticable. The second is whether people understand their retained decisions and can intervene effectively. The third is whether scope and exposure remain bounded when work crosses agents, queues, and service boundaries. None is answered by the conceptual specification alone.

A study should compare at least four conditions on the same tasks and infrastructure: human approval for each consequential effect; static scoped permissions with ordinary audit logs; decision-class ladder labels without the full lifecycle and composition rules; and the proposed grant-based arrangement. The third condition tests whether the extra machinery adds value beyond labels. The first is not presumed optimal: frequent approvals may impose enough load to undermine meaningful review [12, 13].

Tasks should include a communication workflow, reversible operational remediation, and multi-agent research or processing. Use synthetic or consented data in an isolated environment. Independently define an authorization oracle before running the systems: permitted effects, required approvals, reserve categories, cumulative limits, and expected handling of ambiguous cases. Domain experts and affected-party representatives should help specify that oracle where the task requires their judgment. Disagreement must be reported rather than hidden as agent error.

8 2 Fault injection and measurements

Inject changes of recipient, scope, data classification, policy version, and sponsorship; exhausted aggregate budgets; unavailable evidence storage; stale queued authority; duplicate retries; and hostile instructions embedded in retrieved content. Include concurrent child requests, simultaneous reuse of one approval, crashes after dispatch, budget reductions below existing exposure, attempted ledger resets, and revocation during in-flight work. Record both attempted and committed effects so that a model's unsafe proposal is not confused with a system's unsafe external commitment.

Primary measures are unauthorized committed effects per opportunity, severity-weighted exposure, and the upper bound of post-revocation commitments. Secondary measures include task completion, unnecessary escalation, time to detect a boundary violation, review delay, interruption success, evidence completeness, and recovery quality. For rare failures, report uncertainty and exposure counts; observing zero failures is not proof of zero risk.

Human evaluation should test whether reviewers can identify the principal, current authority, material consequences, and required escalation from the presented record. Measure decision accuracy, time, workload, disagreement, and willingness and practical ability to refuse. Include periods of high volume and high apparent automation reliability. A click rate alone cannot validate oversight.

Costs should include engineering, credential isolation, policy maintenance, evidence storage, review labor, drills, and delayed useful work. This is the empirical counterpart of the book's insistence that principles have costs [1, pp. 52–54]. An improvement in one metric should not conceal worsening outcomes for a particular group or loss of contestability.

8 3 Reproducibility and acceptance criteria

Before evaluation, publish or internally freeze the task definitions, policy versions, threat model, randomization procedure, trial counts, scoring rubric, and analysis plan. Use separate development and evaluation scenarios. Record model and tool versions, prompts relevant to the task, seeds where supported, and the exact mediation architecture. Report unavailable controls rather than simulating their existence in the conclusions.

The companion fixture expectations are recorded before their local execution, but were developed alongside the reference model. They are not an independently adjudicated or preregistered research oracle. The comparative study must freeze a separate held-out task set, oracle and

analysis plan before evaluation; fitting the model to its development fixtures cannot validate the claimed organizational benefit.

The initial engineering gate should require every deterministic authorization fixture to satisfy its expected invariant, including concurrent-budget and stale-revocation cases. Passing fixtures is necessary for implementation consistency, not sufficient for deployment safety. Subsequent acceptance thresholds depend on domain consequence, the incumbent workflow, and affected parties; this paper supplies no universal acceptable unauthorized-action rate.

This revision supplies a small executable reference model, a structural JSON schema, machine-readable scenarios with predeclared expected decisions, an example grant, and a reproducible local test report. These are artifacts for review and falsification. They do not instantiate real identity issuance, cryptographic approval verification, durable distributed logs, production endpoint mediation, or an organization capable of meaningful oversight. Independent implementation/security review, finite-state analysis where appropriate, and the four-condition human-and-system evaluation remain future work.

9 Related work and contribution boundaries

9 1 Automation levels and delegation

Sheridan and Verplank's supervisory-control work predates contemporary agents and provides a clear antecedent for degrees of human and computer control [2]. Parasuraman, Sheridan, and Wickens distinguish automation of information acquisition, analysis, decision and action selection, and action implementation [3]. This distinction directly supports the caution against treating every function of an agent as one global autonomy level. Fassio's ladder emphasizes the organizational contract around a consequential decision, including human obligations and reserved governing authority; it should be read alongside these earlier decompositions.

Castelfranchi and Falcone analyze delegation in agent-based systems, including goals, autonomy, control, and forms of delegation [4]. The broad problem of one agent acting for another therefore cannot be attributed to the arrival of language-model systems. This paper's contribution is the source-specific operationalization described in Section 1, with a human sponsorship convention and explicitly retained human authority over the bounds.

Appelo's seven-level management model distinguishes Tell, Sell, Consult, Agree, Advise, Inquire, and Delegate [16]. It is a substantial naming and conceptual antecedent to any present use of delegation levels. Its negotiation of managerial decision involvement differs from this paper's specific, decision-scoped interpretation of Fassio's five published rungs and human-retained governing authority. This distinction identifies the formulation under discussion; it does not establish historical priority or independent validation of it.

9 2 Authorization provenance and current AI agents

Saltzer and Schroeder provide foundational principles of protection, including least privilege and complete mediation [5]. The grant and gateway design relies on that lineage. W3C PROV-DM supplies an established vocabulary for agents, activities, and delegation, including acting on behalf of another party [6]. Such provenance can express relationships but does not itself enforce permission or establish legal responsibility. This paper adds no competing claim to have invented those mechanisms.

Authenticated Delegation and Authorized AI Agents addresses identity, authorization, restricted delegation, and supporting technical representations [7]. Intelligent AI Delegation covers task decomposition, authority, responsibility, monitoring, trust, and adaptive delegation, making it especially close contemporary work [8]. The latter's first preprint version is dated 12 February 2026. This paper does not infer priority over either work from the book's edition year or file creation date.

The grant should be implemented as a profile over existing identity, credential and policy mechanisms where possible. OAuth 2.0 Token Exchange distinguishes delegation and impersonation and represents an actor through the act claim [17]. Its nested actor history is informational, not an automatic conjunction of ancestor business policies, and revocation propagation is not automatic. Those gaps must be resolved by an implementation of this profile rather than attributed to the presence of a delegation token. OAuth Rich Authorization Requests already provides structured `authorization_details` for fine-grained requests [22]; this paper does not claim that existing authorization systems are restricted to coarse scopes.

DPoP binds token use to possession of a key and addresses replay under its protocol assumptions [18]. Mutual-TLS client authentication and certificate-bound access tokens provide complementary identity and token-use controls [19]. SPIFFE supplies workload identities and verifiable identity documents [20]. These mechanisms can authenticate the requesting software and constrain token use. They do not, by their mere presence, establish an accepted human sponsor, a legitimate purpose, shared exposure, or a functioning review contract.

Policy-as-code can express many of the mechanical restrictions proposed here. Open Policy Agent separates policy decisions from enforcement and evaluates structured inputs against policy [21]. A deployment could supply the grant, registry and effect descriptor as policy inputs while an external enforcement point handles reservations and commitment. The contribution sought is therefore a source-traceable governance profile and test protocol for composing existing mechanisms, not a replacement for OAuth, workload identity or policy engines. If a credential-plus-policy deployment already enforces these contracts, shared budgets, lifecycle checks and institutional requirements, the label adds no new protection. The four-condition evaluation must test whether this explicit profile improves implementation and oversight beyond that baseline.

The proposed contribution relative to these works is an explicit correspondence between the published constitutional instruments and a small operational model: five-rung decision classification, one-name sponsorship, human authorization of governing changes, shared exposure, and failure semantics. Several individual mechanisms overlap with established research. Whether their integration yields a practically useful improvement is the subject of the evaluation protocol, not a settled novelty claim.

9 3 Governance and human oversight

NIST AI RMF addresses organizational governance and the mapping, measurement, and management of AI risk [9]. ISO/IEC 42001 describes requirements for an AI management system [10]. IMDA launched its agentic framework on 22 January 2026 and published version 1.5 on 20 May, with the reviewed PDF marked updated 5 June 2026 [11]. It discusses multi-agent cascading failures and coordination risks in Section 1.2.3 (pp. 11–12); least privilege, agent identity and recorded, scoped delegations in Section 2.1.2 (p. 19, pp. 23–24); capable reviewers, oversight audits and denial when approval infrastructure fails in Section 2.2.2 (pp. 29–30); and deterministic system/tool-layer safeguards for higher-risk actions in Section 2.3.1 (pp. 33–34), while retaining a role for model-based safeguards. These controls overlap materially with the present design. The grant schema, numeric examples and revocation semantics are this paper's proposed

representation, not IMDA requirements or evidence of effectiveness. No clause-level conformity or equivalence assessment is offered.

Bainbridge's account of the ironies of automation and Parasuraman and Riley's analysis of use, misuse, disuse, and abuse identify reasons why nominal human involvement can fail [12, 13]. The book's silent-promotion example is consistent with that concern [1, pp. 19–20]. This paper therefore treats review quality and sponsor powers as conditions to investigate, rather than equating human presence with effective control.

Meaningful human control supplies a more demanding comparison than a nominal owner field. Santoni de Sio and van den Hoven distinguish tracking relevant human reasons and environmental facts from tracing system conduct to appropriately informed people who understand its capabilities and their responsibility [23]. The Sponsor of Record convention does not replace those conditions. Its added value, if any, must come from making operational responsibility and intervention testable, while executives and control owners retain their responsibilities.

Oversight can deteriorate under reliable-looking automation. In a simulated multitask experiment, Parasuraman, Molloy and Singh reported poorer failure detection with constant-reliability than variable-reliability automation after approximately twenty minutes; the single-task condition did not show the same deficit [24]. This result motivates the proposed workload and high-apparent-reliability conditions. It does not establish that every reliable agent causes complacency or validate a particular sampling threshold. The present revision checked the study's primary institutional abstract and bibliography, not its inaccessible publisher full text.

For systems classified as high-risk within the EU AI Act's scope, Article 14 addresses oversight proportionate to risk, autonomy and context, including automation-bias awareness, overriding or reversing outputs, and interruption into a safe state [25]. Article 26(2) addresses the competence, training, authority and support of designated overseers. These are relevant legal-adjacent requirements, not duties that can be assumed to apply identically to every agent. A deployment must determine applicability and current obligations through qualified advice. The proposed profile neither supplies that legal determination nor creates a safe harbor, certification or proof of compliance.

Finally, naming does not establish ownership of a general idea. The phrase Ladder of Delegation appears in earlier management writing, including Emery's 2005 essay [15]. Accurate attribution here is to Fassio's book, its specific formulation, and the technical extensions explicitly identified in this paper. It is not an exclusive claim over ladder terminology or every staged-delegation model.

10 Limitations and unresolved choices

The specification remains incomplete as a production architecture. Its companion is an inspectable local reference model, not a credential protocol, policy engine, distributed transaction system, or production log-integrity mechanism. Its strongest conditional invariants require complete mediation, sound scope evaluation, and cross-system ordering that the companion does not demonstrate. Partial adoption must report weaker guarantees precisely. Structural validation, fixture success, and a retained sponsor name cannot be promoted into a claim of deployment safety.

Natural-language intent cannot in general be compiled into a complete set of mechanical predicates. A grant may be internally consistent and still pursue a harmful or illegitimate objective. Human authors can omit a constraint, approve an unjust policy, misunderstand evidence, or place

a sponsor under pressure. The model makes some of these choices visible but cannot make them right.

One named sponsor improves findability only if the surrounding organization gives that person adequate authority, resources, and protection. It can otherwise concentrate blame on a person while leaving actual power elsewhere. The sponsor convention must coexist with clear responsibilities for deployers, vendors, managers, reviewers, and governing bodies. It cannot be used to immunize them from accountability.

The relationship between Rungs 3 and 4 requires domain-specific interpretation. Both allow machine choice within bounds; their difference depends on the actual supervision and review contract. This paper specifies that contract rather than inventing a universal numerical threshold between them. The later 0–5 operating scale introduces another axis and must not be silently substituted for the original ladder.

Reversibility is also contextual and incomplete. Recovery mechanisms may restore service while failing to repair social, economic, or privacy harms. Some decisions require human presence or a legitimate appeal process regardless of technical rollback. The abstract model cannot establish those relationships through an API response.

The book's fiduciary, professional, and maritime analogies motivate retained responsibility [1, pp. 48–51]. This paper does not adopt them as a universal statement of applicable law. Whether duties exist, who bears them, and what remedies are available depends on jurisdiction, role, contract, and facts. Operational sponsorship creates a governance record; it does not by itself create a license, insurance policy, fiduciary relationship, or legal safe harbor.

The executed evidence consists of finite synthetic fixtures, unit checks, and one deliberately restricted differential baseline. It is not a benchmark of deployed authorization products, a field trial, a human study, a machine-checked proof, or a measured safety improvement. These development checks cannot establish how real sponsors behave or how external services commit. Independent technical review and comparative evaluation remain necessary before stronger effectiveness claims.

11 Conclusion

The Constitution of the Frontier supplies a normative architecture for keeping delegated judgment answerable to humans. Its five-rung Delegation Ladder is one instrument within a larger structure that includes expressed intent, earned scope, practical sponsorship, termination, disclosure, and reserved human decisions. Preserving that structure matters: a global agent score or a renamed ladder can obscure the obligations the book places on the delegator.

This revision makes the proposed operationalization inspectable through versioned grants, effect-level admission, action-bound approvals, shared exposure, explicit review and sponsorship obligations, and a small executable companion. The conditional invariants explain what the modeled transitions preserve when their assumptions hold. The evaluation protocol, rather than a claim of proven safety, is the contribution ready for others to scrutinize: compare authorization arrangements under a fixed oracle and test both committed effects and practical human intervention. Wider empirical and independent review remain necessary. The objective is accountable delegation with demonstrable retained human control.

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Appendix A Source traceability and terminology

A1 Complete constitutional principle names

The following names preserve the completed first-edition structure [1, pp. 58–59]. They are book-origin principles, not newly introduced technical requirements.

Article I The License

- 1 Intent is the license for delegation
- 2 Every delegation names its principal

Article II The Limits

- 3 Autonomy cannot exceed traceability
- 4 Autonomy is earned, not configured
- 5 Verify against irreversibility
- 6 Authority moves by decision, not by drift

Article III The Conduct

- 7 Use judgment inside boundaries
- 8 Disclose to the person, not to the file
- 9 Watch the gap between the measure and the mission
- 10 Leave a trail someone actually reads

Article IV The Accountability

- 11 Responsibility has to be able to land somewhere
- 12 Every autonomous actor has a human sponsor of record
- 13 Nothing runs that can't be stopped

Article V The Reserve

- 14 Some decisions stay human
- 15 Reserve the decisions where we'd want the rule broken

The meta-rules are “Every principle must cost something” and “No principle may rest on what machines can't do” [1, p. 58]. The book's interpretation and qualifications remain important; this list alone is not a substitute for them.

A2 Source to extension ledger

Table 5 Published foundations and proposed extensions

Concept	Published source	Added in this paper
Intent Contract	Chapter 3 pp. 25–27 and Appendix B p. 60	Version binding and linkage to enforceable scope predicates
Delegation Ladder	Chapter 2 p. 19 and Appendix B p. 60	Typed decision instances and rejection of numeric permission ordering
Explicit authority changes	Chapter 2 p. 22	Immutable grant versions and activation transitions
Trust Economy	Chapter 4 pp. 29–31 and Appendix B pp. 60–61	Shared reservation ledger and bound approval records

Concept	Published source	Added in this paper
Human reserve	Chapter 5 pp. 34–37 and Appendix A p. 59	Human-governed register of reserved decisions and roles
Agent Personnel File	Chapter 6 pp. 39–40 and Appendix B p. 61	Links among actors, grants, capabilities, dependencies, and evidence
Sponsor of Record	Chapter 6 pp. 40–41 and Appendix A p. 59	Registry semantics, accepted succession, and historical attribution
Termination	Chapter 6 p. 41 and Appendix B p. 61	Queue handling, revocation fencing, and residual-effect accounting
Named principal and lineage	Chapter 8 pp. 48–51	Explicit separation of principal, issuer, sponsor, and approver
Complete constitution	Epilogue pp. 52–54 and Appendix A pp. 58–59	Selective technical interpretation with organizational limits retained
Later chain controls	Agentic Futures pp. 8 and 14	Explicit restriction semantics, atomic exposure reservation, and revocation ordering assumptions

The title “Delegation Ladder” follows Appendix B. “Delegated Judgment” is the title of Chapter 2. This paper does not claim a new exclusive name by combining those phrases. Sponsor of Record, Intent Contract, Trust Economy, and Agent Personnel File are already present in the published book. Architectural requirements for constrained chains, cumulative budgets, and revocation are present in the later white paper. The particular grant schema, admission predicates, approval-binding semantics, atomic-ledger model, conditional invariants, reference model and evaluation protocol are explicit technical extensions proposed here.

A3 Source and authorship record

The controlling book source is the complete first-edition manuscript identified by Ed Fassio as the published edition. The reviewed PDF contains 66 physical pages, including four preliminary pages and 62 numbered pages; references use the printed pagination. Its edition year is 2026. This source record does not establish a more precise public release date. The later Agentic Futures source is the 18-page revised edition dated 2 October 2026, distinguished from the earlier public download.

Ed Fassio is the sole named author. Generative AI assisted source comparison, literature discovery, drafting, technical critique, reference-model and fixture authoring, and document preparation. AI assistance is not listed as coauthorship. The technical extensions and synthetic development checks are identified separately from the published book. Independent technical review and empirical evaluation remain outstanding.

Appendix B Later operating model extensions

B1 Two different scales

The revised Agentic Futures working paper uses six levels numbered 0 through 5: Human Only, Recommend, Prepare, Act Within Limits, Manage Outcomes, and Adaptive Delegation [14, p. 6]. Those labels belong to a later professional-services formulation. The published book uses five rungs numbered 1 through 5, ending in human-retained Set the bounds. The two scales differ in both numbering and emphasis.

Table 6 is an interpretive crosswalk proposed in this paper. It is not a translation that preserves level numbers. The later scale emphasizes how work is prepared and orchestrated; the book asks who holds the consequential choice and who governs its bounds. Determining the book rung requires examining each decision in the workflow.

Table 6 Conditional crosswalk for the later six level formulation

Later level	Possible book interpretation	Required qualification
0 Human Only	Human choosing and acting; information support may be present	No dedicated matching book rung for the whole workflow
1 Recommend	Rung 2 for the proposed consequential decision	A human must actually decide before commitment
2 Prepare	Rung 1 for specified preparation or Rung 2 for a proposal	Preparation may itself have protected effects; execution needs its own authority
3 Act Within Limits	Rung 1 for fixed human choices or Rung 3 for machine choices	Determine who chooses; thresholds alone do not settle the rung
4 Manage Outcomes	Rung 3 or Rung 4 depending on decision and review arrangements	Human-drawn bounds and required human decisions remain in force
5 Adaptive Delegation	A workflow can combine Rungs 1–4 and constrained descendants	Adaptive coordination never gives a machine the book's Rung 5 governing authority

This separation also resolves a potential ambiguity in “adaptive.” A machine can vary the route through an already authorized space without changing the boundaries of that space. Expanding the space, removing a required approval, or redefining who may authorize it is a governing change. The latter remains human in the book-based model, regardless of the complexity of the orchestration.

B2 Four labels as a compression

Intent, Boundaries, Legibility, and Human Sponsorship in the later paper summarize important themes [14, p. 5]. They omit distinctions that are explicit in the completed book, including naming the principal, disclosure and recourse, proxy drift, terminability, reserved human presence, and mercy. Future summaries should label the four-part presentation as a compression or adaptation rather than state that the book itself consists of four principles.

The later paper's organizational-owner language also requires care. The book's Sponsor of Record is a named human with practical powers and explicit succession. A department or function can provide support but does not substitute for that named human within the book's convention. The present specification retains the stricter source definition.

Appendix C Research roadmap

The next milestone is independent examination of the specification and executable model, followed by a preregistered comparative study. The existing local development checks make selected rules inspectable; they do not establish the organizational effectiveness of the framework.

C1 Independent technical examination

Reviewers should examine the grant schema, R1–R7 admission conditions, state transitions, approval binding, shared exposure, revocation ordering, and implementation assumptions. They should test whether the worked traces and failure responses follow from the stated contracts, identify missing transitions, and challenge trusted inputs. Finite-state exploration where appropriate and a separately implemented reference model would help expose errors that authored fixtures can miss. Record findings and corrections in subsequent reports.

Source-focused review should examine the mapping from the book's five rungs and four instruments to the proposed technical extensions, especially Rungs 3 and 4, human-only governing authority, retained decisions, sponsorship coverage and reversibility. The later Agentic Futures 0–5 scale must remain separately identified. Challenge contribution boundaries through a broader literature review.

C2 Preregistered comparative evaluation

The primary study should follow Section 8 with four conditions: human approval for each consequential effect; static scoped permissions with ordinary audit logs; decision-class ladder labels without the full lifecycle and composition rules; and the proposed grant-based arrangement. Use the same tasks and infrastructure, with an independently adjudicated authorization oracle, held-out evaluation scenarios and a frozen analysis plan. Report disagreements, excluded cases and uncertainty rather than optimizing the oracle after observing results.

Compare attempted and committed effects separately. Test the four focal failures, stale authority and distributed-system faults against declared endpoint assumptions. Measure unauthorized commitments, missed authorized work, escalation burden, review delay, retained exposure and operational cost. Report weaker mediation guarantees.

The pinned development companion also outlines simpler mechanism-level comparisons. Those suggestions and its isolated differential fixture are preparatory probes, not substitutes for the four-condition study defined here.

C3 Human control in practice

A human study must examine whether sponsors and reviewers understand the active mandate, can inspect relevant evidence, have adequate time and competence, and can halt or refuse work without pressure from conflicting performance incentives. Include absence and succession, high workload, apparently reliable automation, escalation to specialist control owners, and actual cessation drills. A recorded name, approval click or available button is insufficient evidence of effective control.

Evaluate review quality, recovery and residual harms, with appropriate consent and institutional safeguards. Distinguish software behavior, organizational support and normative choices.

C4 Evidence and versioned refinement

Report artifact versions, dependencies, tasks, oracle decisions and limitations. Revise the specification in response to counterexamples, preserving the distinction between book-origin claims, proposed extensions and empirical findings. A repository or persistent identifier would improve citability, not establish peer review or validation.